



Glass as hard as Diamond

China has developed a glassy material hard and strong enough to scratch diamonds. <https://digit.in/sep21-19>



Real Dairy, no cows involved

California's Perfect Day uses fungi to make dairy protein that is identical to the protein in cow's milk. <https://digit.in/sep21-20>

LIVING MATERIALS

Your smartphones could soon have living cells inside them.

Satvik Pandey | feedback@digit.in

We tend to think of the things around us as either living or non living things, but what if they could just be both. Researchers are breaking down this barrier by creating things that are both living and non living at the same time. When you think about it there are bits and pieces of us that are not really living, such as nails and hair, but living materials does not refer to just the stuff of life., but rather a novel type of materials where both kinds of matter are fused. These materials are unique because they have laid the foundations for electronics being made from living cells. These electronics and material incorporate microorganisms

within them to give them unique properties and capabilities. To put it in a simple sentence, scientists have been able to combine and program living and non-living materials in a way that they can predict and control the development of the living part of the material to make it more diverse in terms of its properties and the dynamic changes that the non-living part and living part together can go through. The first concept of living materials in its nascent stage dates back to 2014 when a team of bright minds from MIT had developed living materials for the first time.

Timothy Lu, an assistant professor of electrical engineering and biological engineering and an author of the paper that described these newly found materials at the time had said, "Our idea is to put the living and the nonliving worlds together to make hybrid materials that have living cells in them and are functional... It's an interesting way of thinking about materials synthesis, which is very different from what people do now, which is usually a top-down approach."

At the time, what Lu and his team had developed was groundbreaking and set the foundations of the research and development that is going on today. Lu and his team had worked with the bacterium E. Coli because it produces amyloid proteins that help E. coli attach to surfaces, called 'curli fibres'. Also E. Coli are like the lab rats of the microbial world. These were made of repeating chains of identical proteins and could be modified by adding peptides. The peptides that had been introduced into the systems could attach themselves to non-living materials, which, as used in the experiments, were gold nanoparticles. But the products that resulted from their experiments were not at all practical for large scale uses. In 2021, Lu's thoughts and ideas have started to become a reality. Along with Tom Ellis, who is a professor of bioengineering at Imperial College, London, he has made this advancement. It uses some traits of kombucha tea and its fermentation process to make cellulose embedded with enzymes. These newly developed